

RAJDEEP GILL

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EDUCATION

Bachelor of Science in Computer Engineering (Co-op)

University of Manitoba

GPA: 4.45/4.5

Relevant Coursework: Data Structures and Algorithms, Systems Engineering Principles, Parallel Programming

May 2026

Winnipeg, MB

EXPERIENCE

Undergraduate Research Student

May 2024 - Dec 2024

Biosensors Research Lab - University of Manitoba

Winnipeg, MB

- Designed and trained a convolutional neural network to predict biological cell viability from 10,000+ diffraction pattern images, achieving 85% accuracy and demonstrating the potential for non-invasive cell viability assessment.
- Enhanced particle diffraction centroid detection using edge detection and clustering algorithms, improving success rate by 17% (from 75% to 92%) and providing more accurate data for the CNN model.
- Developed a Python-based data visualization tool with an interactive GUI to process, filter, and display diffraction patterns and model predictions, enabling researchers to analyze and interpret data efficiently.
- Designed and implemented a scalable SQL database to centralize 100,000+ optical diffraction records, replacing a manual file-sharing system.

Undergraduate Research Student

May 2023 - Aug 2023

Biosensors Research Lab - University of Manitoba

Winnipeg, MB

- Applied dimensionality reduction techniques (t-SNE, PCA) to analyze and cluster datasets, uncovering key correlations that informed future research directions and improved data interpretability.
- Optimized data analysis workflows by implementing parallel processing techniques, reducing processing time by 90%.
- Developed an oscilloscope interfacing program to automate data collection, improving speed and accuracy.
- Delivered weekly progress updates, communicating findings and incorporating feedback to refine research approaches.

PROJECTS

Google Docs Clone | React, TypeScript, HTML/CSS, Next.js

- Developed a Google Docs clone with real-time collaboration, user authentication, and document sharing.
- Utilized Convex to store and sync document data, allowing multiple users to edit the same document simultaneously.
- Implemented organization-wide document sharing and user permissions to control access to documents.

Financial Dashboard | React, TypeScript, HTML/CSS, Next.js, Hono.js

- Created an expense and income tracking dashboard with user authentication and real-time data visualization.
- Implemented a RESTful API to handle user data and transactions, and a PostgreSQL database to store user information.

Discord Clone | React, TypeScript, HTML/CSS, Next.js

- Developed a Discord clone with user authentication, real-time messaging, video and voice chat functionality.
- Utilized Next.js for server-side rendering, socket.io for real-time messaging with a polling fallback.

Microprocessor Communication System | C, Java, Android Studio

- Designed and implemented a server using the TCP/IP stack in C, handling multiple concurrent connections.
- Developed a Java-based proxy server to interface between an Android application and the microprocessor server.
- Designed an Android application to allow users to control the LEDs, monitor temperature and button states.

PUBLICATIONS

B. Arzhang, E. Kovacs, J. Lee, **R. Gill**, E. Salimi, D. Thomson, and G. Bridges, "Machine Learning Classification of Microplastics by Integrating Optical and Dielectrophoresis Features," in *IEEE AP-S/URSI* 2025.

B. Arzhang, G. Bridges, **R. Gill**, J. Lee, E. Kovacs, E. Salimi, and D. Thomson, "Dielectric Analysis of Single Biological Cells at Both Microwave and Optical Frequencies," in *IEEE AP-S/URSI* 2025.

TECHNICAL SKILLS

Languages: Python, Java, C, C++, JavaScript, TypeScript, HTML/CSS, R, SQL, MATLAB

Frameworks & Libraries: React, Next.js, TensorFlow, PyTorch, OpenCV,

Developer Tools: Git, VSCode, Visual Studio, Android Studio, IntelliJ